

# ■ Deep Thought — AI Agent Assignment

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Use Case: AI Exam Revision Assistant

## ■ Objective

To design an AI agent that helps students revise for their exams by breaking down the learning process into four key thinking steps — understanding, tracking, planning, and generating helpful responses.

## ■ 4-Step Architecture Design

| Layer                  | Prompt                                                                                              | Example Output                                                                                                                                                                                                                                                      |
|------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Input Understanding | Understand what the user is asking. Identify the subject and chapter to focus on from the question. | User: 'Help me with science from the question chapter do you have?'<br>AI: 'I understand. You want to revise the Electricity chapter, correct?'<br>User: 'Yes, please.'<br>AI: 'Great! Let's start with Ohm's Law — it states that Current = Voltage / Resistance.' |
| 2. State Tracker       | Track which chapters are completed or pending.                                                      | AI: 'You've completed Motion and Light are done, and Electricity is in progress.'                                                                                                                                                                                   |
| 3. Task Planner        | Plan what the agent should do next based on the user's state.                                       | AI: 'Since you're revising Electricity, I'll generate 5 questions for you to practice.'                                                                                                                                                                             |
| 4. Output Generator    | Generate clear, motivational, and structured responses.                                             | AI: 'Good! Let's start with Ohm's Law — it states that Current = Voltage / Resistance.'                                                                                                                                                                             |

## ■ ChatGPT Exploration Log

| Attempt | Prompt Variant                                 | What Happened                       | What Changed                    | Why                             |
|---------|------------------------------------------------|-------------------------------------|---------------------------------|---------------------------------|
| 1       | Generate 5 revision questions for Electricity. | Output was too generic.             | Added chapter context and tone. | To make it more personalized.   |
| 2       | Generate conceptual + numerical questions.     | Questions were repetitive.          | Added instruction for variety.  | To improve engagement.          |
| 3       | Act as a friendly tutor                        | Perfect output — clear and helpful. | Kept as final version.          | Achieved best clarity and tone. |

## ■ Output Tests

**\*\*Test 1 – Normal Input:\*\***  
Input: 'I want to revise the Electricity chapter.'  
Output: 'Sure! Let's start with Ohm's Law — Current = Voltage / Resistance.'

**\*\*Test 2 – Vague Input:\*\***  
Input: 'Help me with science.'  
Output: 'Which chapter would you like to study — Motion, Light, or Electricity?'

**\*\*Test 3 – Invalid Input:\*\***  
Input: '' (empty)  
Output: 'Please enter a topic or subject you need help with.'

## ■ Reflection

1. The hardest part was getting the AI to retain chapter memory correctly.

2. I enjoyed crafting prompts that made the AI sound like a friendly teacher.
3. If given more time, I would add visual summaries or diagrams for each topic.
4. I learned how prompt clarity changes AI performance drastically.
5. When I got stuck, I tested smaller prompts and observed the differences.

### ■ Hack Value (Optional Exploration)

I explored adding memory tracking for chapters and used structured prompts to simulate student progress. This helped the agent act more like a learning assistant rather than a static chatbot.

### ■ Conclusion

This project demonstrates that an AI can be guided to think step-by-step — transforming a simple chatbot into a truly helpful learning companion through modular prompt design and structured reasoning.